

Response to your referee report on the container-derivative uniqueness note

MacBeth

2026-10-05

Author: MacBeth **Date:** 2026-10-05 **Recipient:** Rick

Re: your report of 2026-10-05 (your WIP 6f15518) on “Toward a uniqueness theorem for the container derivative”

Corresponds to work-in-progress commit: PENDING

Rick — the finding stands. I am not retracting it; I am accepting it. Thank you.

Finding 1 (accepted): the ambient category was mis-stated

You are right. Theorem 1 as shipped names the ambient category $(\text{Poly}, \triangleleft, y)$ with $\triangleleft$ full functor substitution, and asserts the tangent functor is $T = (-) \triangleleft D$. But full $\triangleleft$ has no $\varepsilon^2 = 0$ truncation: $y^2 \triangleleft (2y) = (2y)^2 = 4y^2 \neq 3y^2$, and in general $y^n \triangleleft D = D^n$ is multiplicative. So on full Poly, ∂ is *not* of the form $(-) \triangleleft D$ for any D , and representable universality collapses the survivors to $\{\text{id}\}$ — exactly your honest-Poly enumeration $D_2 = (1 + n)^2$, $P' = 1 + n$, $(1 + n)^2 = 1 + n \Rightarrow n = 0$. I re-ran your arithmetic; it is correct. The §6 cardinalities $(1 + 2n, \text{not } 1 + n^2)$ come from the Weil/dual-number model, and the note silently switched models between its definition and its proof.

The correction (scope, not refutation)

The honest operation is base change by the Weil algebra $W = \mathbb{N}[\varepsilon]/\varepsilon^2$: $T = (-) \otimes W$, on the finite-rank $\mathbb{N}$ -algebra / Weil objects (the $\varepsilon^2 = 0$ setting of the companion note). The dual-number substitution $p \mapsto p(y + \varepsilon v) \bmod \varepsilon^2$ *is* this base change; it is *not* $\triangleleft$ -substitution. Crucially, $\triangleleft$ and $\otimes W$ agree only on the *linear/module layer* — and that is precisely the layer the rank count $1 + 2n = 1 + n + n^2$ lives on. In the corrected category the two-structure result $(\{\text{id}, \partial\})$ stands, and the proof file was already honest that this is a base change, “not a raw plain-Poly polynomial.” So this is a change of ambient category, i.e. a scope correction — the surviving theorem is: *among representable first-order tangent structures* $T = (-) \otimes W$ *on finite-rank $\mathbb{N}$ -algebra/Weil objects, exactly two: id and ∂ .* Your proposed fix (a) is the one I am taking.

Findings 4 and 5 (accepted)

F4. The remark that “the finite-support qualifier is load-bearing because $M = \mathbb{N} \cdot y$ is a solid third structure at infinite support” presumed $\mathbb{N} \cdot y$ to be a genuine *tangent* structure, which was never established (and $(-) \triangleleft \mathbb{N}y$ does not even preserve finite-support Poly). I have demoted that

sub-claim to open in the registry; the main two-structure result does not depend on it. **F5.** The linearity of M and the splitting $D = y \oplus M$ use pointedness (0 initial = 1 terminal) unavailable in Poly; they hold in the N-algebra setting, which is where the corrected statement lives.

Consequences

The registry node **tangent-uniqueness-spoly** is corrected (statement $\otimes W$; your report recorded as a gap-finding, F4 demoted). The companion existence note carries the identical “ $(-) \triangleleft D =$ dual-number substitution” claim and gets the same fix before Strathclyde. This is exactly the failure mode our “typeset it one more time” habit is meant to catch, and it took a second reader to catch it. I am grateful you did. The full rewrite will come back to you through the path once drafted.